

TED(10) - 4069

MODEL QUESTION PAPER (REV-10)
FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY
(Branch MD only)
MEDICAL IMPLANTS AND ASSISTING DEVICES

(Maximum Marks: 100)

TIME: 3HRS

PART - A

Answer all Questions in one or two sentences. Each question carries **2 marks**.

I.

1. Define bio-compatibility?
2. Define prosthetic devices?.
3. What the basic principle of fibre –optic dializer?.
4. Differentiate PEEP and NEEP?
5. What you mean by Tidal volume? **5x2=10 marks**

PART - B

Answer any **FIVE** of the following questions. Each question carries **6 marks**.

II.

1. List out the basic properties of biomaterials? Different kind of biomaterials and give examples?
2. Describe artificial heart valve or mechanical heart valve?
3. Explain any two membranes for haemo dialysis machine?.
4. Differentiate spontaneous and forced ventilation?
5. Compare homograft and heterograft?
6. What are polymers? explain bio-degradable polymers?
7. Explain basic principle of artificial kidney?

5x6=30 marks

PART - C

Answer one full question from each Unit. Each question carries 15 marks.

UNIT I

III a) Explain material response, tissue response, host response to biomaterial? (8).

b) Explain acrylic polymers and give examples? (7)

OR

IV. a) Categorise ceramic materials? (8)

b) Explain polyoxymethylenes and its applications? (7)

UNIT II

V. With neat diagram explain the design consideration of hip prosthesis? (15)

OR

VI a) Describe crutches and canes? (12)

b) Describe biological graft (3)

UNIT III

VII a). With neat schematic, explain the basic working principle of heart lung machine? (10)

b) List out any two types of oxygenators? (5)

OR

VIII a) Explain the block diagram and operation of hemodialysis machine (15)

UNIT IV

IX . a) Define ventilation? Explain schematically positive pressure ventilation? (9)

b) What are the consignments required for manual artificial ventilation? With diagram (6)

OR

X a) Explain the block diagram of micro processor controlled ventilator. (12)

b) Differentiate anesthesia ventilator and intensive care ventilators (3)

(4 X 15 = 60)